Today's Agenda

(1) House keeping:

- Test 2
- Assignment 2

(2) Addressing Queries

(3) Ecology, Unit 2

(4) Biotic and Abiotic Factors

(5) Energy flow in the ecosystem

Introduction

- The Namibian in 2019 carried an article titled **"Stranded whale euthanised at Walvis Bay"**
- The humpback whale got stranded on WB's Independence Beach.
- For several days, about 8 attempts were made to redirect the whale into deeper ocean, but it kept coming back to the shore.
- Efforts were made to push it back into deeper waters hoping that it would decide to swim back into the Atlantic ocean and fight to stay alive
- All efforts could not avail
- People mobbed on the struggling whale for photos
- The whale was eventually had to be euthanised.

UNIT 2

Ecology

Learning Outcomes

By the end of this unit students should be able to:

- *explain* the essentials of ecology, some ecological components, principles and terminology;
- *explain* how all factors, living and non-living, are dependent on each other and are in constant interaction with one another;
- *explain* what an ecosystem is, and how we can study one;
- *summarise* the concept of energy flow through a food web.

Define Ecology

- The study of the interactions that take place among organisms and their environment

includes all abiotic and biotic factors in one particular environment

Biotic Factors

the living parts of an ecosystem

Abiotic Factors

the nonliving parts of an ecosystem

Examples of Abiotic Factors







include air, water, soil, light, temperature, wind, source of energy (usually sun)

Examples of Biotic Factors

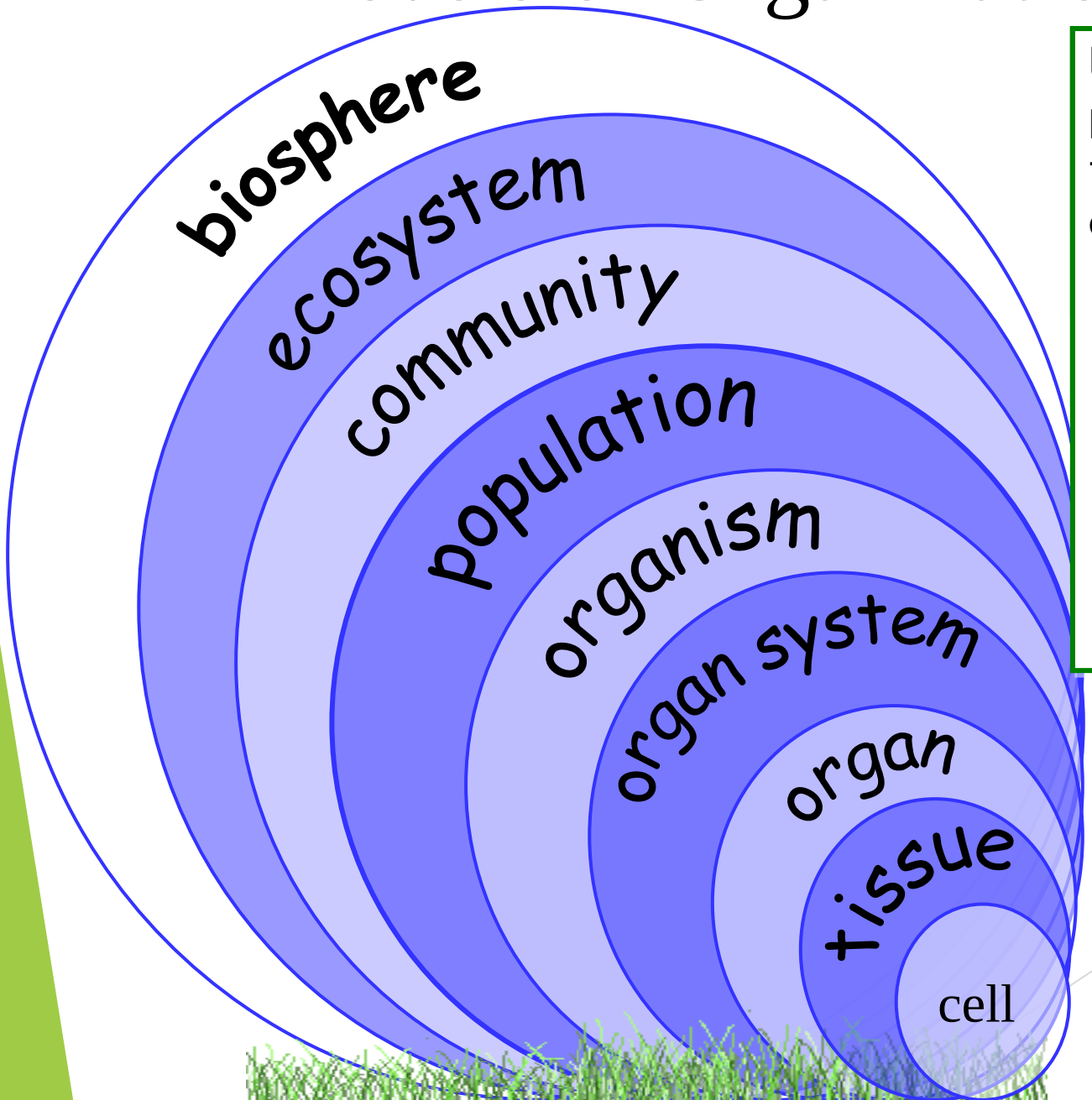






include plants, animals, fungi, microorganisms

Levels of Organization



large region with typical plants and animals that includes several ecosystems

Group Discussion

- In groups of 3-4 students, discuss what you understand by the following ecological terms.

(a) Organism

(b) Population

(c) Species

(d) Community

(e) Ecosystem

(f) Biosphere

What is a Organism?

One individual living thing



What is a Population?

- A group of individual of the **same species** living in a particular area.
- For example, all foxes living in an area form a population.
- Another example, all dandelions growing in an area form another population.
- **Population ecology** analyzes factors that affect population size and how and why it changes through time.





Species

- Species - a group of organisms which can interbreed and produce fertile offspring

What is a Community?

- A community is all the populations of different species living in the same area.
- All the foxes, grasshoppers, snakes, hawks, and deer living in one area each form their individual populations, but together make up a community.
- **Community ecology** examines how interactions between species, such as predation and competition, affect community structure and organization.



Ecosystem

Community + Abiotic environment, interacting

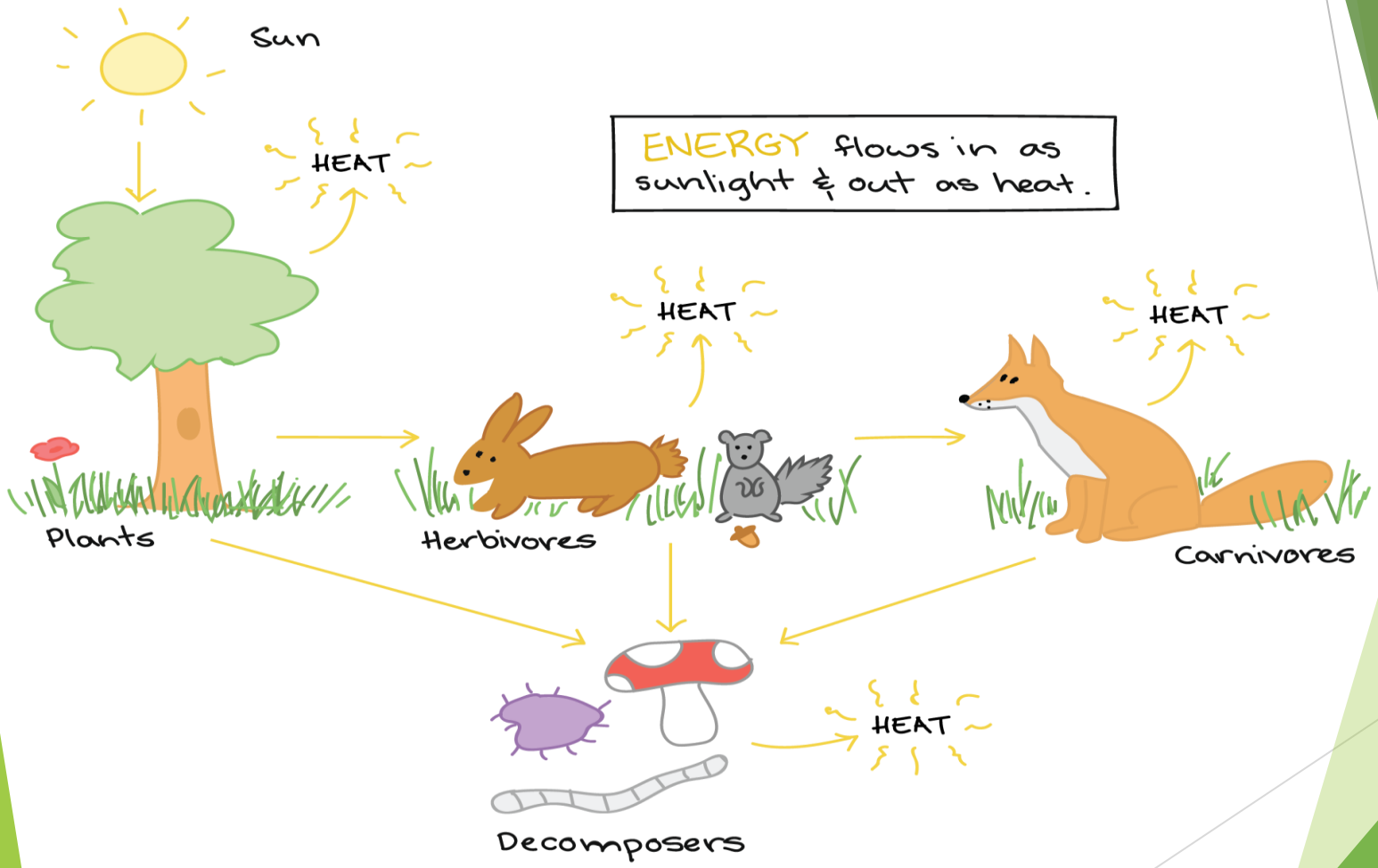


Groups of living things interact within ecosystems

The environment can be organized into five levels

1. **Biosphere : region with similar climate, types of** plants, and animals
2. **Ecosystem: The living and non- living things that** interact in one environment.
3. **Community:** *All living populations* **of the different species** *found in an area.*
4. **Population: A group of organisms of the same species** that live in the same area.
5. **Organism: A single living thing, made up of one or many cells,** that is capable of growing and reproducing.

Energy Flow in the ecosystem



The biotic components

- All the **living factors** in an ecosystem are known as the biotic components.
- The biotic components of an ecosystem can be divided into **two large groups**
- **Autotrophic and Heterotrophic**
- the **autotrophic component** (self feeding – make their own food) and the **heterotrophic** component (various feeding)
- Communities contain autotrophs and heterotrophs

Energy source

- All organisms require a source of energy to live.
- Sunlight is the sources of energy that powers almost all life processes on earth.

Autotrophs

- Autotrophs are also called **Producers** because they produce all of the food that heterotrophs use
- Without autotrophs, there would be no life on this planet
- Ex. Plants



Producer

- Most producers are autotrophs
- and use light as an **energy source** for manufacturing organic compound from carbon dioxide and water
- **Sunlight, water** and **carbon dioxide** into **Glucose** (food) through the process called photosynthesis.

Heterotrophs

- Organisms that do not make their own food
- Another term for Heterotroph is **consumer** because they consume other organisms in order to live
- "Consume" means "eat".
- Animals are **consumers** because they "eat" (consume) food provided by plants or other animals.

- Ex. Rabbits, Deer, Mushrooms



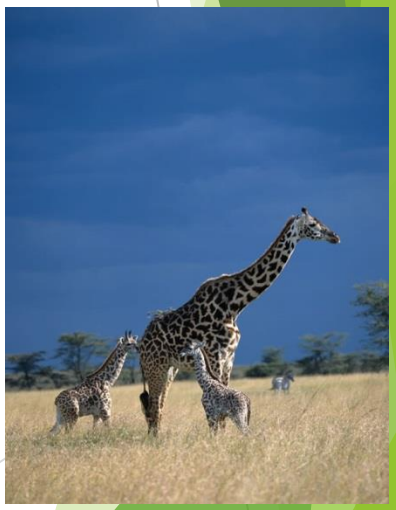
Heterotrophs

Primary Consumers

1. **Herbivores** – eat ONLY plants
   - Ex. – Cows, Elephants, Giraffes







Heterotrophs

Secondary Consumers

2. Carnivores – eat ONLY meat

- Ex. – Lions, Tigers, Sharks





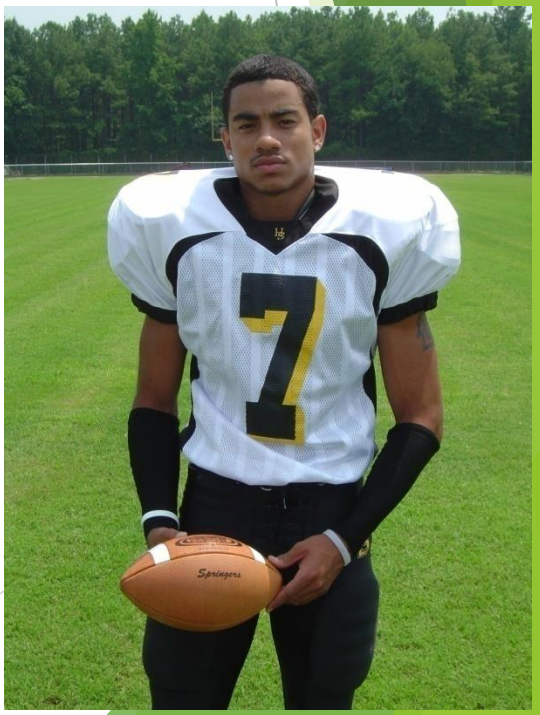


Heterotrophs

Consumers

3. **Omnivores** – eat BOTH plants and animals
   - Ex. – Bears and Humans





Heterotrophs

Consumers

4. **Scavengers/Detritivores** – feed on the tissue of dead organisms (both plants and animals)
   - Ex. – Vultures, Crows, and Shrimp

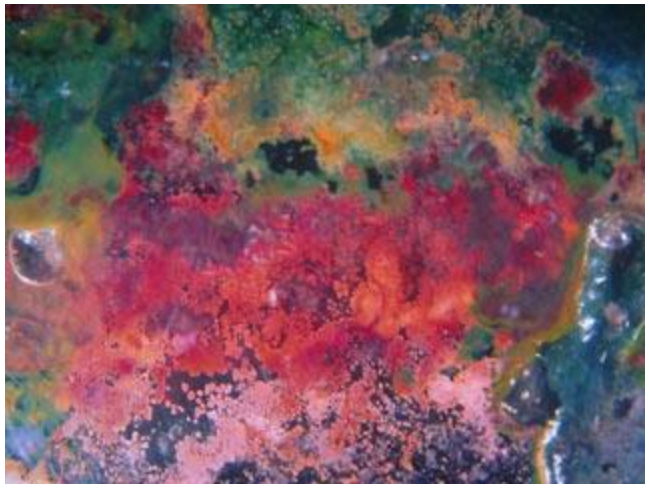






Heterotrophs

5. **Decomposers** – absorb any dead material and break it down into simple nutrients or fertilizers
   - Ex. – Bacteria and Fungi





Decomposers break down nutrients in the dead "stuff" and return it to the soil.

The producers can then use the nutrients and elements once it's in the soil.

The decomposers complete the system, returning essential molecules to the producers.

1. Ecologically speaking, grasses are classified as _____ and deer are classified as_____
2. Name two groups are absolutely essential to the functioning of an ecosystem?
3. In a tropical rain forest, the producers obtain energy from the_____
4. What do ‘self-feeding’ autotrophs require from the environment in order to make their own food?
5. Even consumers at the highest level of a food web eventually become food for _____

1. Silver maples, red oaks, red wolves, deer, robins, bluejays, a fresh water stream, river otters, black rat snakes, various soil bacteria and fungi, loamy soil, and lots of rain in the spring could describe a(n) _____.
2. Ecology is the study of _____.
3. When ecologists study the abiotic components of the environment they are studying _____, _____, _____.
4. Green plants are_____
5. All the various species interacting in a location make up a (n) _____.

Today's Agenda

(1) House keeping:

- Assignment & Test

(2) Addressing Queries

(3) Recap of Monday's class

(4) Energy flow in the ecosystem (Food Chain & Food Web)

(5) Endangered and Extinct species

(6) Interaction of organisms in the environment

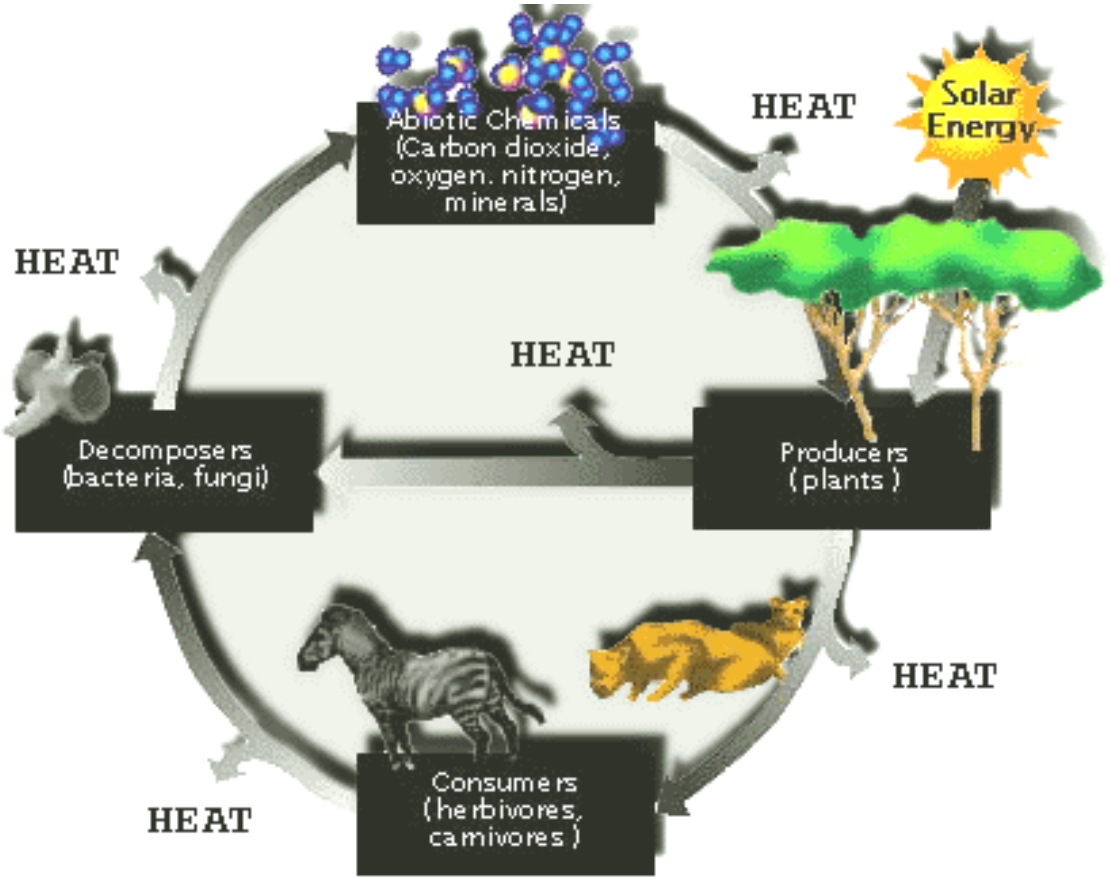
Lesson Outcomes

- By the end of this class *students should be able to:*
- *describe* how the feeding relationships in an ecosystem can be expressed as food chains
- *summarise* the concept of energy flow through a food web
- *explain* how energy losses occur along food chains and discuss the efficiency of energy transfer between trophic levels
- *explain* the application of ecological concepts in everyday living
- *differentiate* between endangered and extinct
- *define* competition and distinguish between interspecific and intraspecific competition
- *distinguish* among mutualism, commensalism, and parasitism, giving examples of each

Introduction

- An article was reported by the Namibian titled **“Traditional Leader Nabbed for Poaching”**
- Three suspects among whom is a traditional leader were arrested for allegedly poaching 3 springboks found in their possession.
- But well all granted bail of N$500 each.
- There was also a reports on “Police, suspected poacher exchange gunfire”

Processes of Ecosystems



Trophic Levels

- Energy moves from one organisms to another when it is eaten

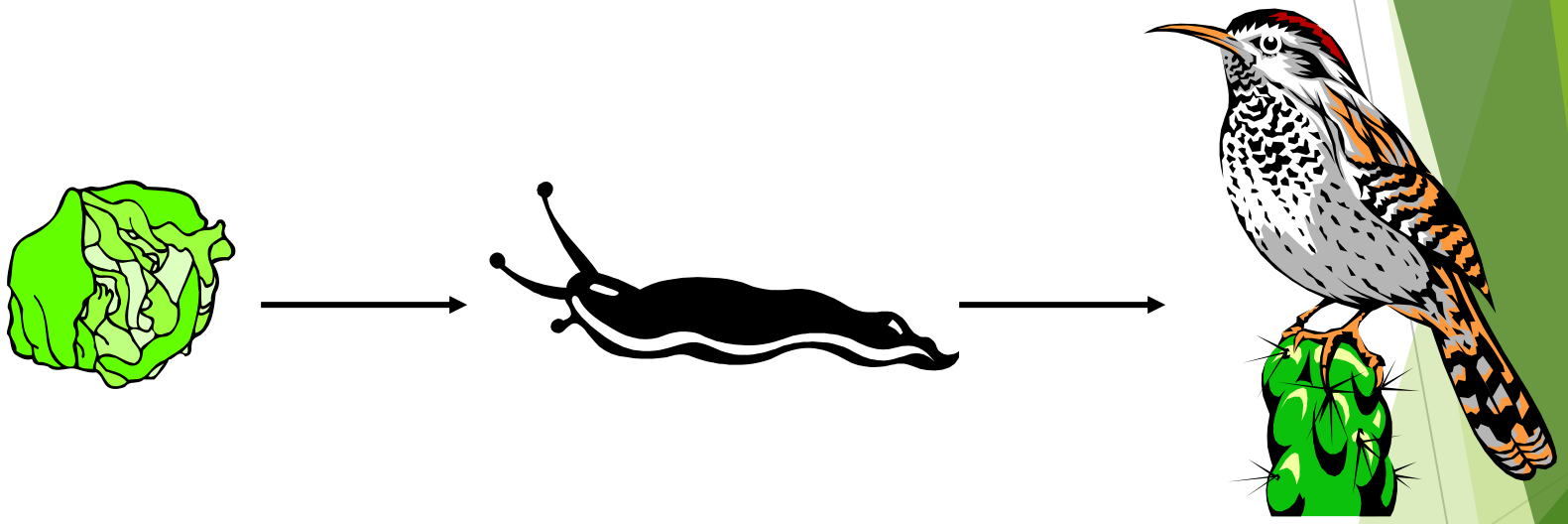
- Each step in this transfer of energy is know as a **trophic level**
  - The main trophic levels are **producers, consumers, and decomposers**

The flow of energy through ecosystems is **linear**. Energy flow occurs in **food chain**.

Food Chains

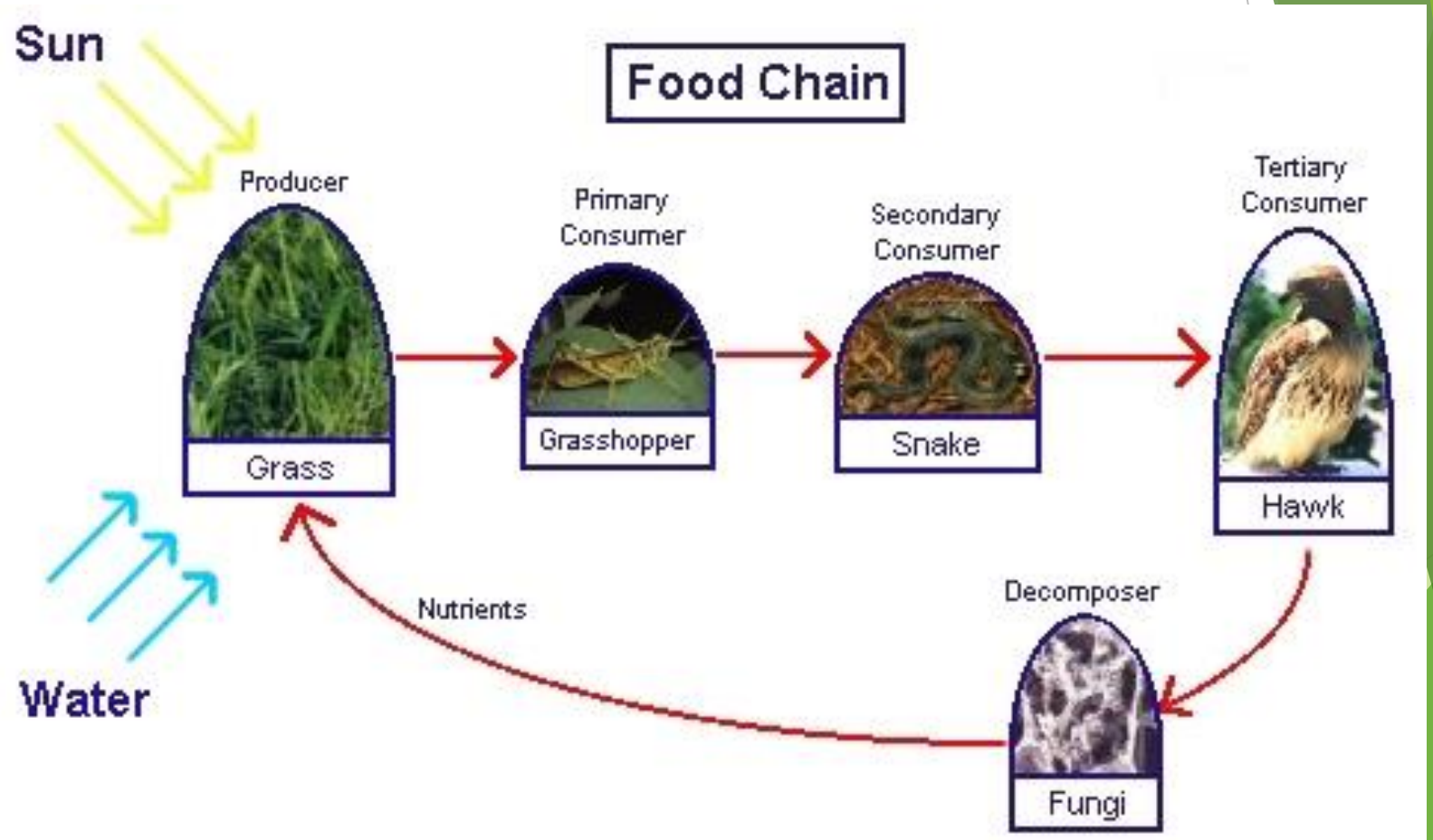
- The energy flow from one trophic level to the other is know as a **food chain**
- A food chain is simple and direct
- It involves one organism at each trophic level
  - Primary Consumers – eat autotrophs (producers)
  - Secondary Consumers – eat the primary consumers
  - Tertiary Consumers – eat the secondary consumers
  - Decomposers – bacteria and fungi that break down dead organisms and recycle the material back into the environment

Food chains always start with a plant.

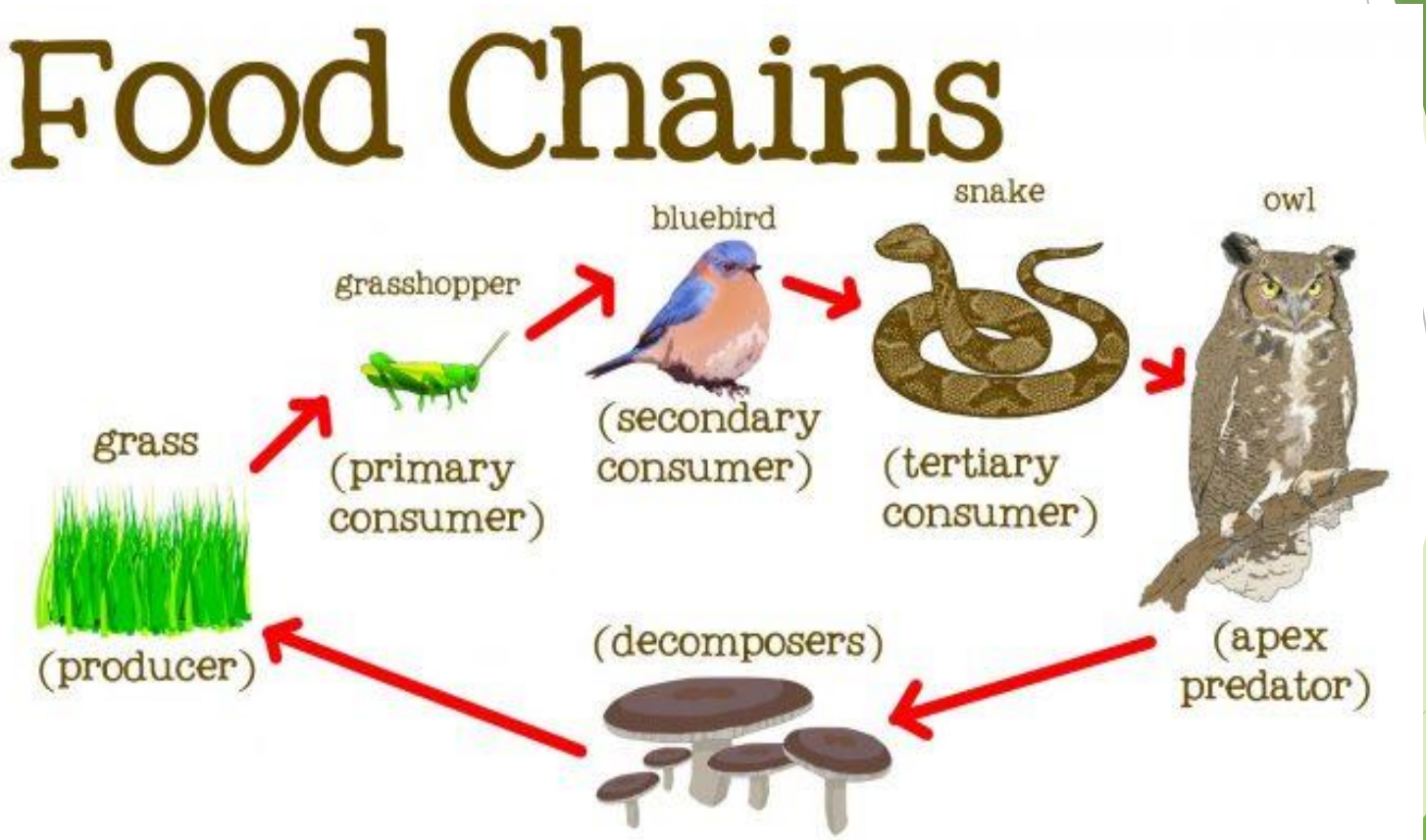


The lettuce is eaten by the slug, the slug is eaten by the bird.

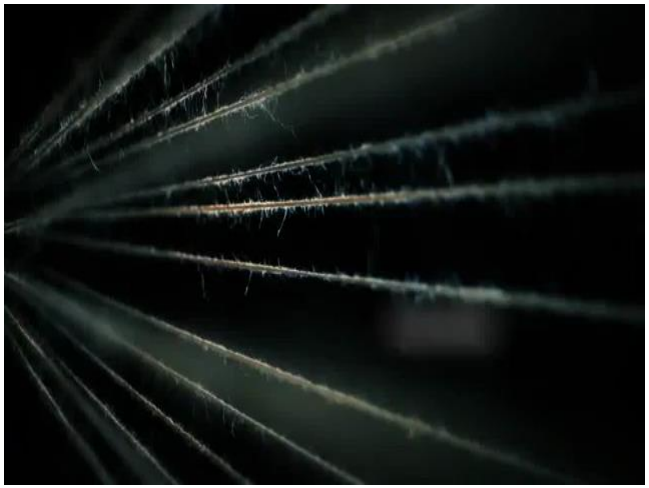
Food Chain


Sun
Food Chain
Producer
Primary Consumer
Secondary Consumer
Tertiary Consumer
Grass
Grasshopper
Snake
Hawk
Nutrients
Decomposer
Water
Fungi

Food Chain


Food Chains
grass
(producer)
grasshopper
(primary consumer)
bluebird
(secondary consumer)
snake
(tertiary consumer)
owl
(apex predator)
(decomposers)

Food chains

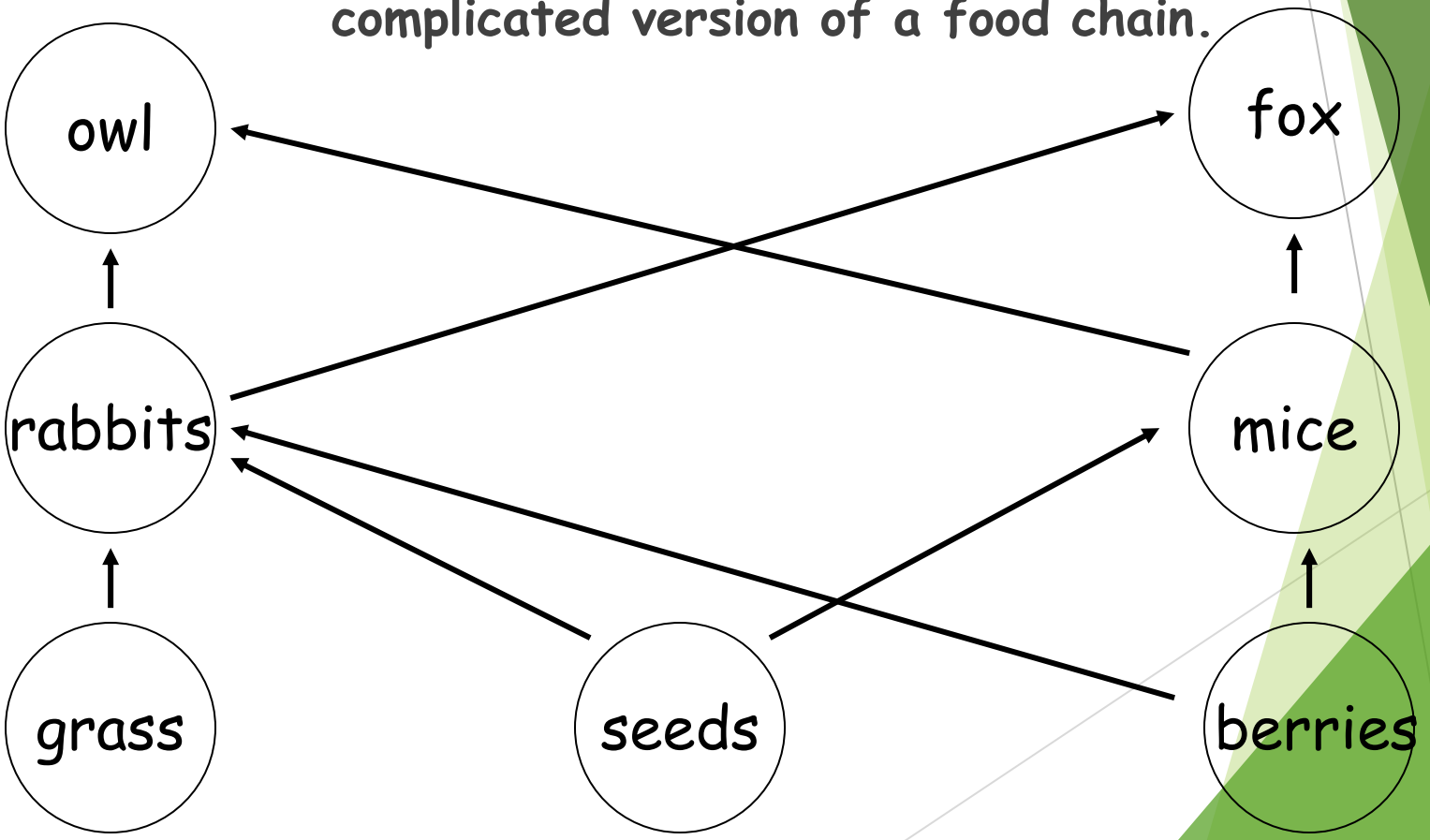


Food Web

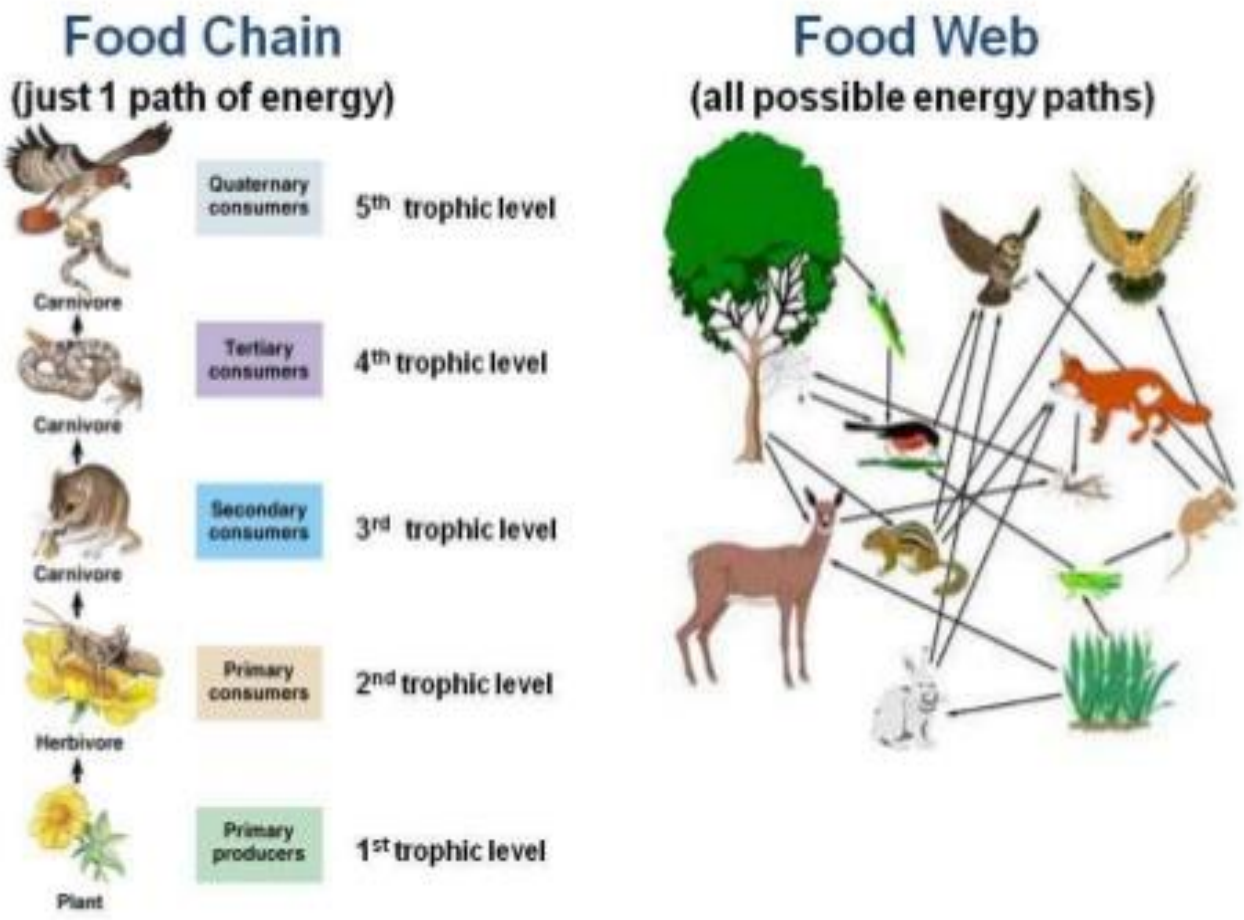
- Most organisms eat more the JUST one organism.
- When more organism are involved it is know as a **FOOD WEB.**
- Food webs are more complex and involve lots of organisms.
- A food web, which is a more complex of interconnected food chains in an ecosystem.

Food Webs

We can show this by using a food web, which is just a more complicated version of a food chain.



Food Chain & Food Web



The *arrow* points to the eater and shows the transfer of energy.

Trophic level

- Think about it for a moment, how does energy flow?

- The most important thing to remember about **energy flow** in ecosystems is that it is **linear**, or **one-way**.
- That is, energy can move along a food web from one trophic level to the next trophic level as long as it is not used to do biological work.
- Once energy has been used by an organism, however, it is lost as heat and is unavailable to any other organism in the ecosystem.

Transfer of Energy

- When a zebra eats the grass, it does not obtain all of the energy the grass has (much of it is not eaten)
- When a lion eats a zebra, it does not get all of the energy from the zebra (much of it is lost as heat)

Transfer of Energy

- The two (2) previous examples of energy transfer show that no organism EVER receives all of the energy from the organism they just ate

- Only 10% of the energy from one trophic level is transferred to the next – this is called the 10% law

Energy loss in a food chain

- Only 10% of energy is passed along to the next level in a food chain. The rest of the energy is lost through:
- **-waste**
- **-heat**
- **-movement**
- **-respiration**



Why is the transfer of energy in an ecosystem referred to as energy flow, not energy cycling?

- Energy passes(flows) through an ecosystem, entering as sunlight, moving as transfers of chemical energy in the food web, and leaving the ecosystem as heat.
- It is not recycled within the ecosystem.

Group discussion

- In Groups of 5-6

Discuss the effect of breaking the chain? For instance what will happen if all the snakes for some reasons dies out complete in a given ecosystem.

- 3-4 minutes

Breaking the Chain

- Organisms living in a habitat depend on each other.
- If one part of a food chain dies out or is greatly reduced, the consumers have to find alternative food, move away, or starve.
- This then affects more consumers in the same way.

Introduction to Endangered & Extinct Species



Endangered or Extinct?

- The number of people in the world is growing at an alarming rate.
- But this is not true for all animals.
- In some cases, there are only a few of one type of animal left in the wild.
- These animals are **endangered**.
- If they die out completely, they become **extinct**.

Interaction time on Endangered/ extinct species

- **What causes species to become endangered/extinct?**

Why does this happen?

There are lots of reasons why animals become endangered or extinct.

The most common are:

- loss of habitat (woodlands cut down, rivers drying up, bush encroachment).
- chemicals or pollution poison the animals.
- hunting (for sport, their fur, tusks or meat or collecting).

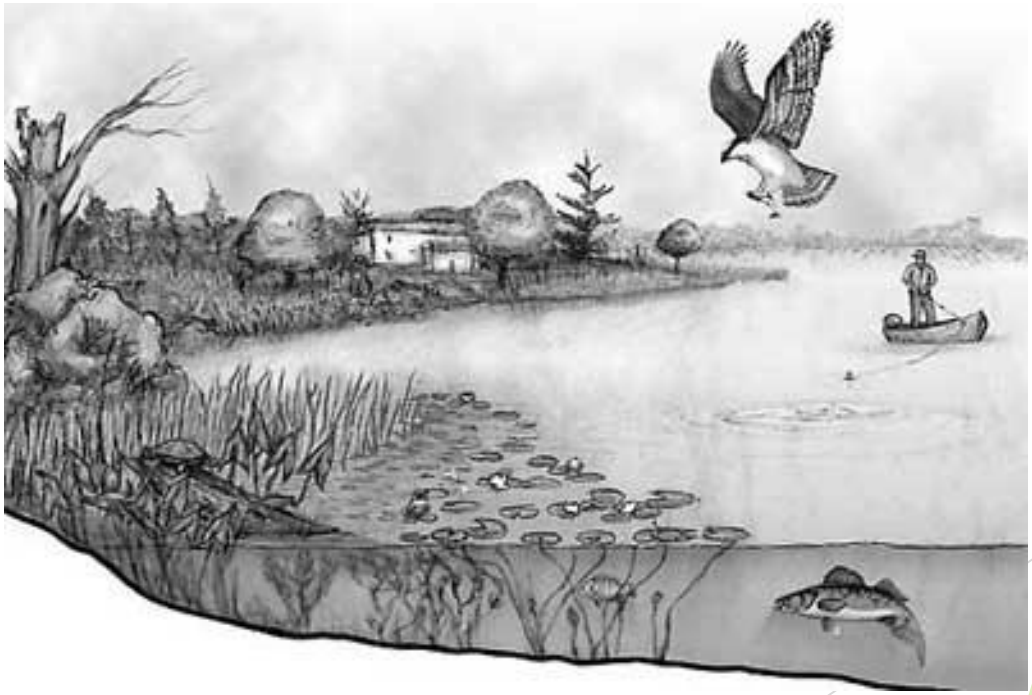
Caring for the Environment

- It is in our own best interests to look after the world we live in.
- If a habitat is lost or damaged, it has an effect on everything else, even if we do not see or understand it straight away.
- Remember - once something becomes extinct, it's gone forever!

Group Discussion

- In groups of 5-6
- As a community member, you are consulted by officials from MEFT seeking the voice of the community on how to cab the prevalent problem of poaching in general. What will be your suggestions?
- 5 minutes

Interactions Within Ecosystems



Types of ecological interactions

Competition

Cooperation

Predation

Symbiosis

- Parasitism
- Mutualism
- Commensalism

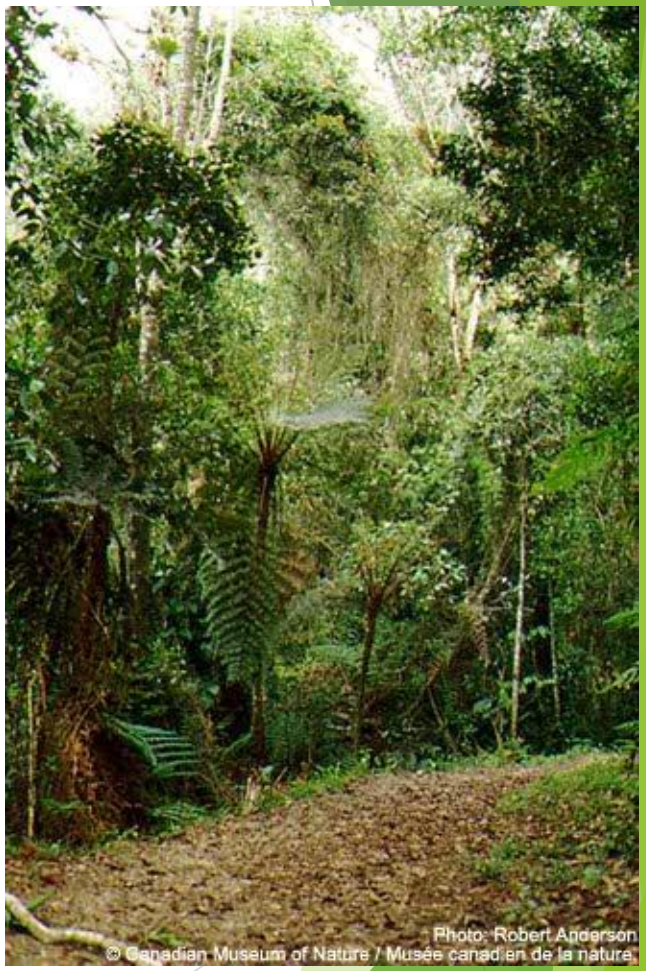
Organisms Interact in Different Ways

Competition

- Competition is the struggle between individuals or different populations for a **limited resource**



- **Intraspecific** - Competition can happen with the same species (plants compete for light, space, and nutrients)
- **Interspecific** - Competition between different species (hyenas and vultures compete for remains of dead animals)



Cooperation



Some organisms work together to benefit each other

- Killer whales hunt in pods (groups)
- Ants, bees, and termites (members of a colony have different roles and responsibilities…queen bee, worker bees, etc.)



Predation – one species (prey) feeds on another (predator)



Symbiosis:

- Two different species who live together in a close relationship
  - Both species benefit
  - One species benefits while the other is not affected
  - One species benefits while the other is harmed

Types of Symbiosis- Mutualism:

- Two species interacting with each other that benefits both species.
- bees and flowers







Types of Symbiosis - Commensalism:

- Two species interacting with each other with **one species benefiting and the other unaffected**.
- jellyfish and fish

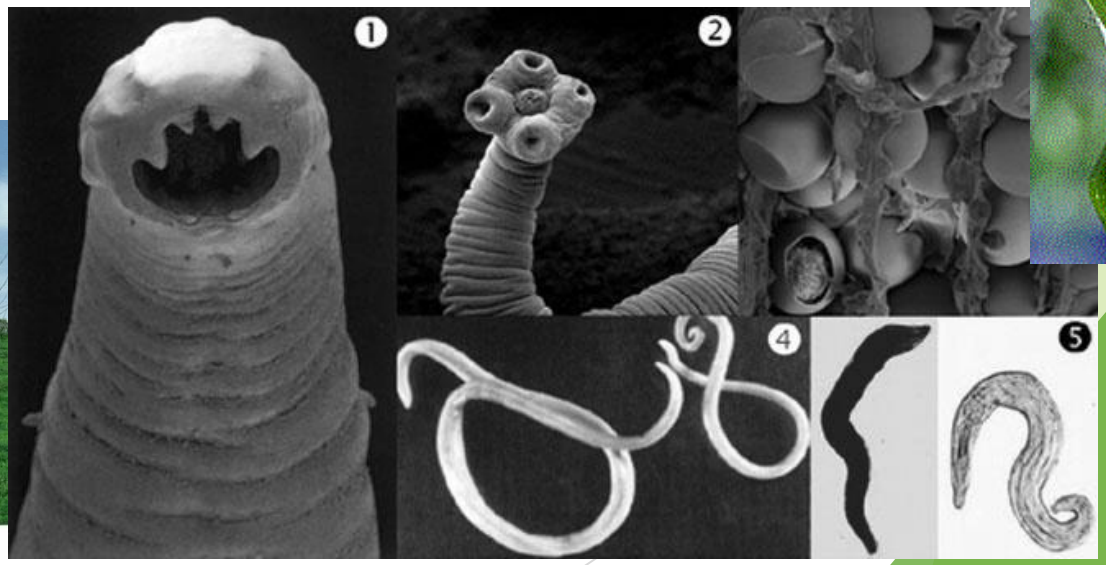


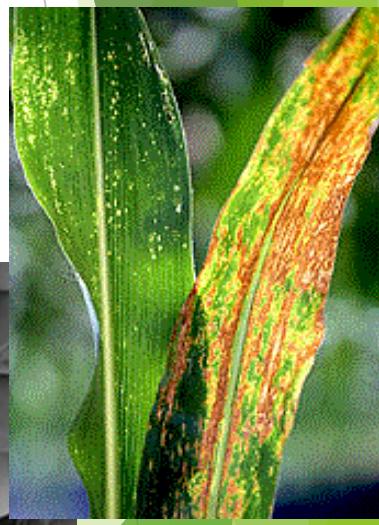


Types of Symbiosis - Parasitism:

- Two species interacting while **one species benefits and the host species is harmed**
- Examples of human parasites







There are two types of parasites:

- The **ectoparasite** **living outside** the body of the host:

  - ticks (parasite) on cattle (host)
  - lice on fowls;
  - fleas on dogs and cats
  - Aphids on plants

- The **endoparasites** **living inside** the body of the host; in the digestive system, in the blood vessels, the tissues, the air passages:
  - bilharzias (schistosoma) lives in the bladder or the intestine of man
  - the plant parasite: dodder, living on Lucerne, sunflowers and mealies.

1. Name the two types of parasite.
2. Define Commensalism.
3. Define Competition
4. A tapeworm lives inside the intestines of a cow and feeds by absorbing food that the cow is digesting. This is an example of_____
5. What type of symbiotic relationship is represented in the picture?



1. What is the distinction between interspecific competition and intraspecific competition?
2. Explain how seed dispersal by animals is an example of mutualism in some cases.

1. How much energy is used at each trophic level?
2. A cow is an example of a _____
3. A complex of interconnected food chains in an ecosystem is called a (an) _____
4. Why is the transfer of energy in an ecosystem referred to as energy flow, not energy cycling?

Summary

- **Ecology:**

  The study of the interactions that take place among organisms and their environment

- **Abiotic Factors:**

  The *non living* things in an environment

- **Biotic Factors:**

  All the living things in an environment

- **Autotrophic:**

  Make their own food, with the help of **sunlight**, **water** and **carbon dioxide**
- **Heterotrophic:**

  Organisms that do not make their own food depend on autotrophs
- **Food Chains:**

  The energy flow from one trophic level to the other

- **Food Web:**

  Most organisms eat more than JUST one organism. When more organisms are involved this is know as a FOOD WEB

- **Producer**

  A photosynthetic green plant or chemosynthetic bacterium, constituting the first trophic level in a food chain; an autotrophic organism.

- **Consumer**

  A heterotrophic organism that ingests other organisms or organic matter in a food chain.

- The general term used to describe intimate associations between pairs of species is **symbiosis**.
- Parasitism is an association between two organisms in which the smaller, the **parasite**, lives either temporarily or permanently in or on the larger, the **host**. The host is harmed by the parasite, but is usually not killed by it.

- **Commensalism** is an association in which one of the two organisms benefits, while the other neither loses nor gains.
- **Mutualism** is an association which benefits both participants.

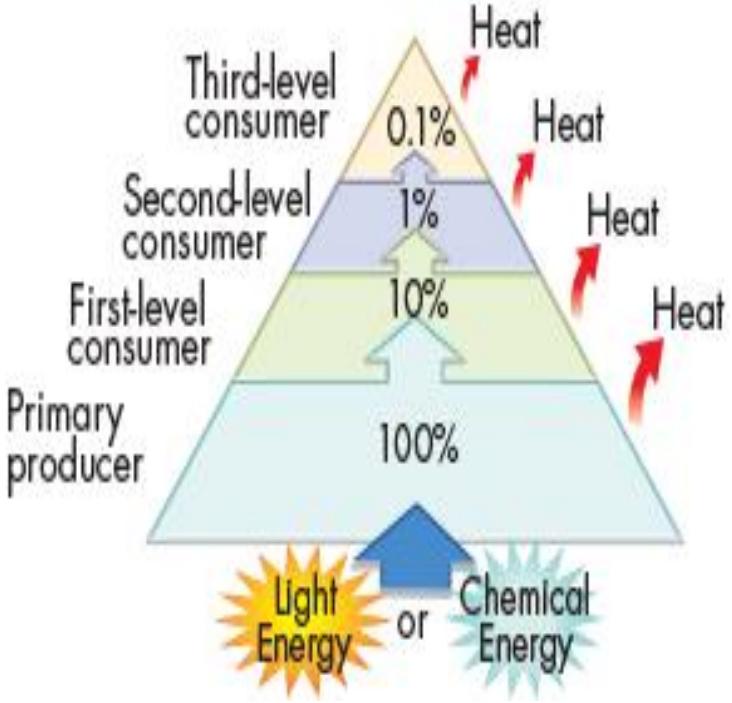
Learning Objectives

- Explain how energy is lost from one trophic level to the other.

- Distinguish between different types of ecological pyramids;
- Pyramid of biomass
- Pyramid of numbers
- Pyramid of energy

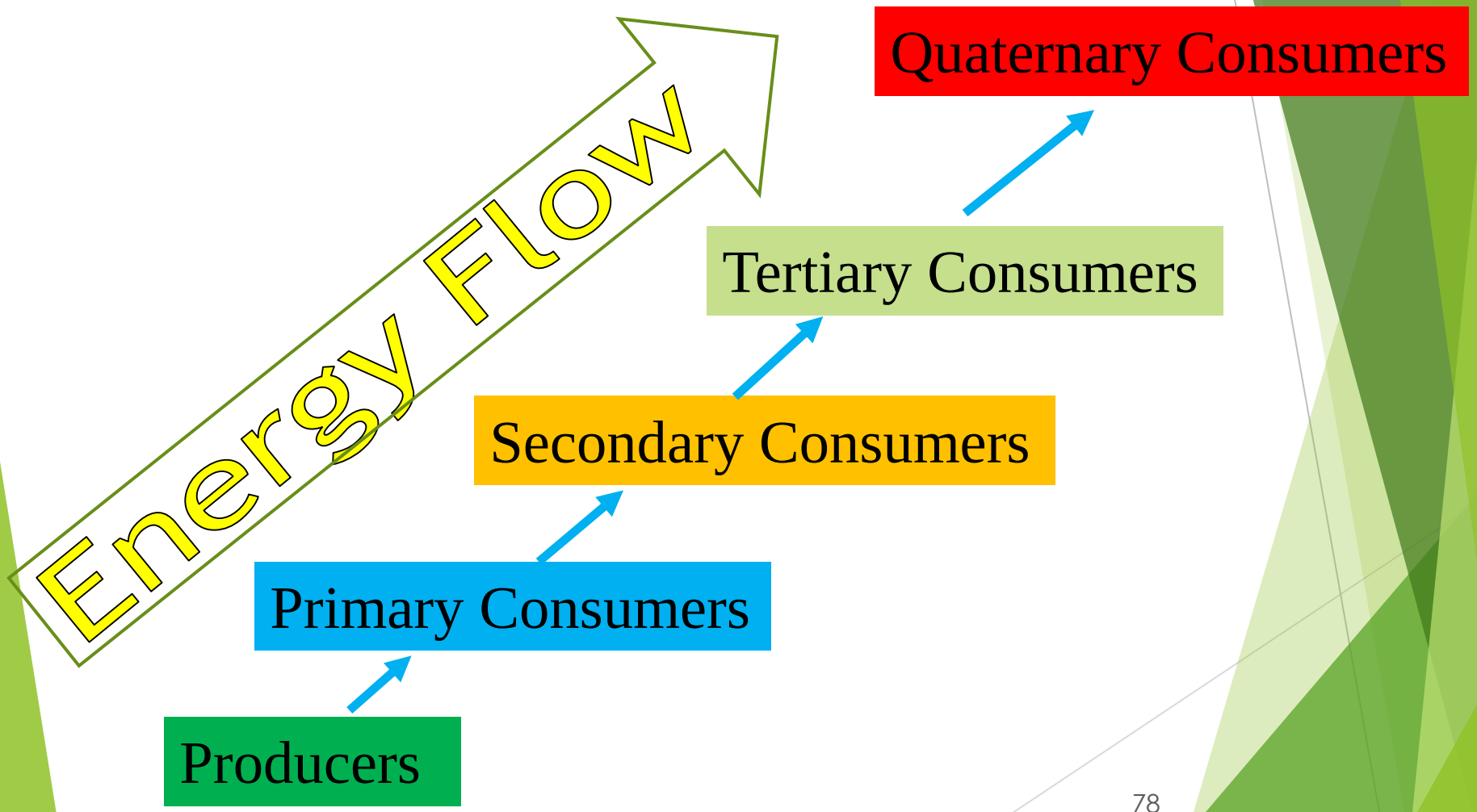
Trophic Levels

- Each step in a food chain or food web is called a **trophic level.**
- Primary producers always make up the first trophic level.
- Various consumers occupy every other level. Some examples are shown.





Trophic Levels



Ecological Pyramids

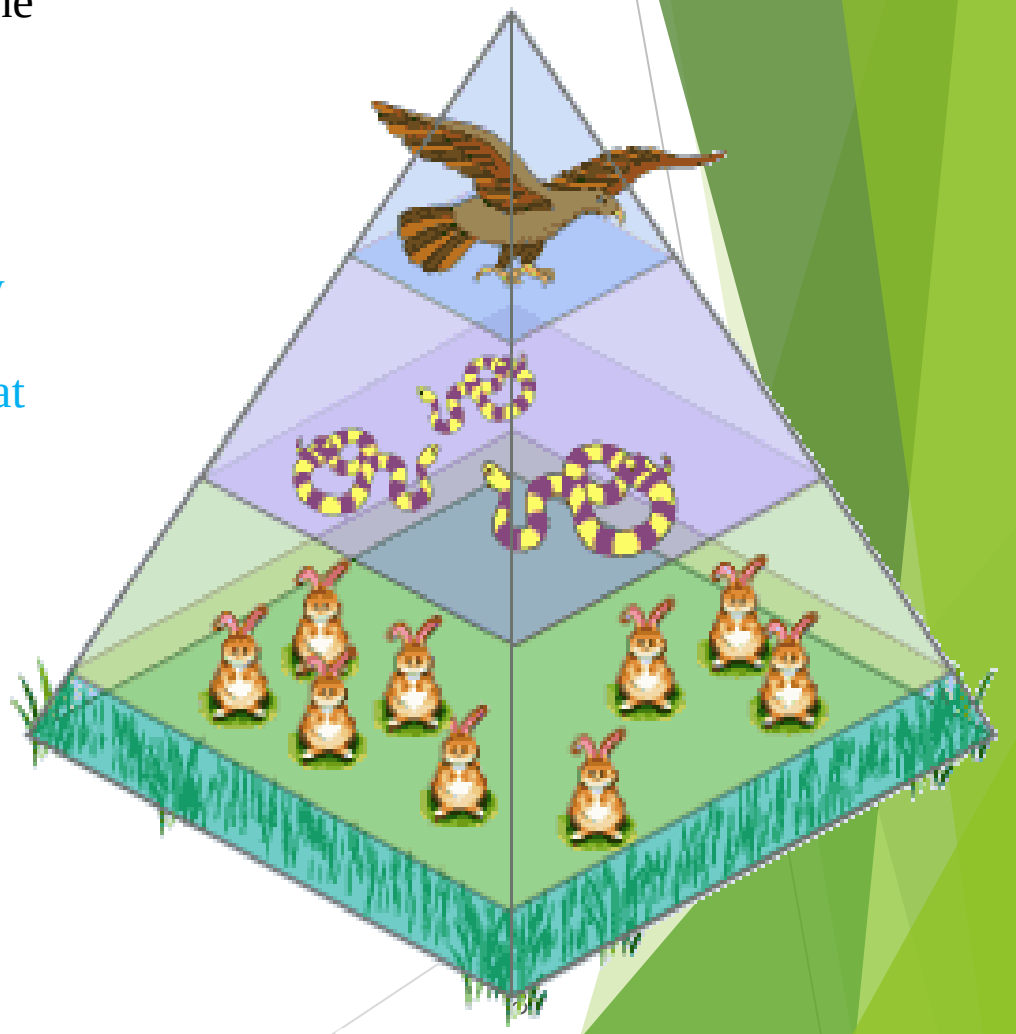
- **Ecological pyramids** show the relative amount of energy or matter contained within each trophic level in a given food chain or food web.
- There are three different types of ecological pyramids:
  - pyramids of energy,
  - pyramids of biomass, and
  - pyramids of numbers

Ecological Pyramids

1. What do the three types of ecological pyramids illustrate?
   - Pyramids of energy show the relative amount of energy available at each trophic level.
   - A pyramid of biomass illustrates the relative amount of living organic matter at each trophic level.
   - A pyramid of numbers shows the relative number of individual organisms at each trophic level in an ecosystem.

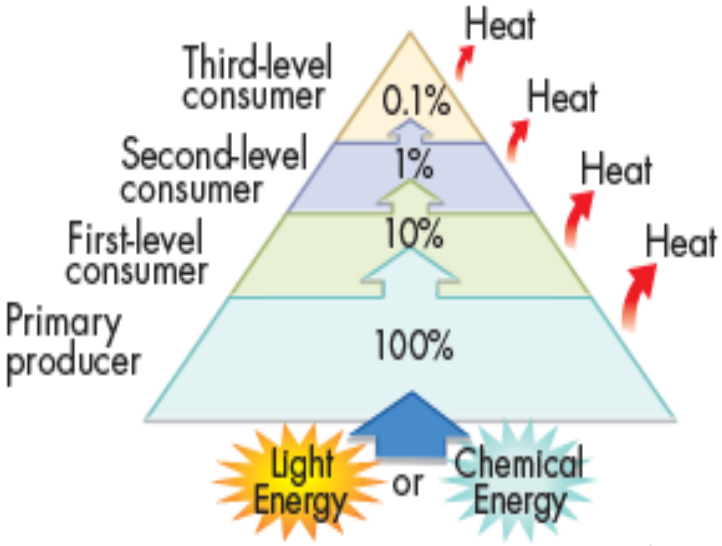
1. Pyramids of Energy

- There is theoretically no limit to the number of trophic levels in a food web or the number of organisms that live on each level.
- However, only a small portion of the energy that passes through any given trophic level is ultimately stored in the bodies of organisms at the next level.
- Organisms expend much of the energy they acquire on life processes, such as respiration, movement, growth, and reproduction.
- Most of the remaining energy is released into the environment as heat—a byproduct of these activities.

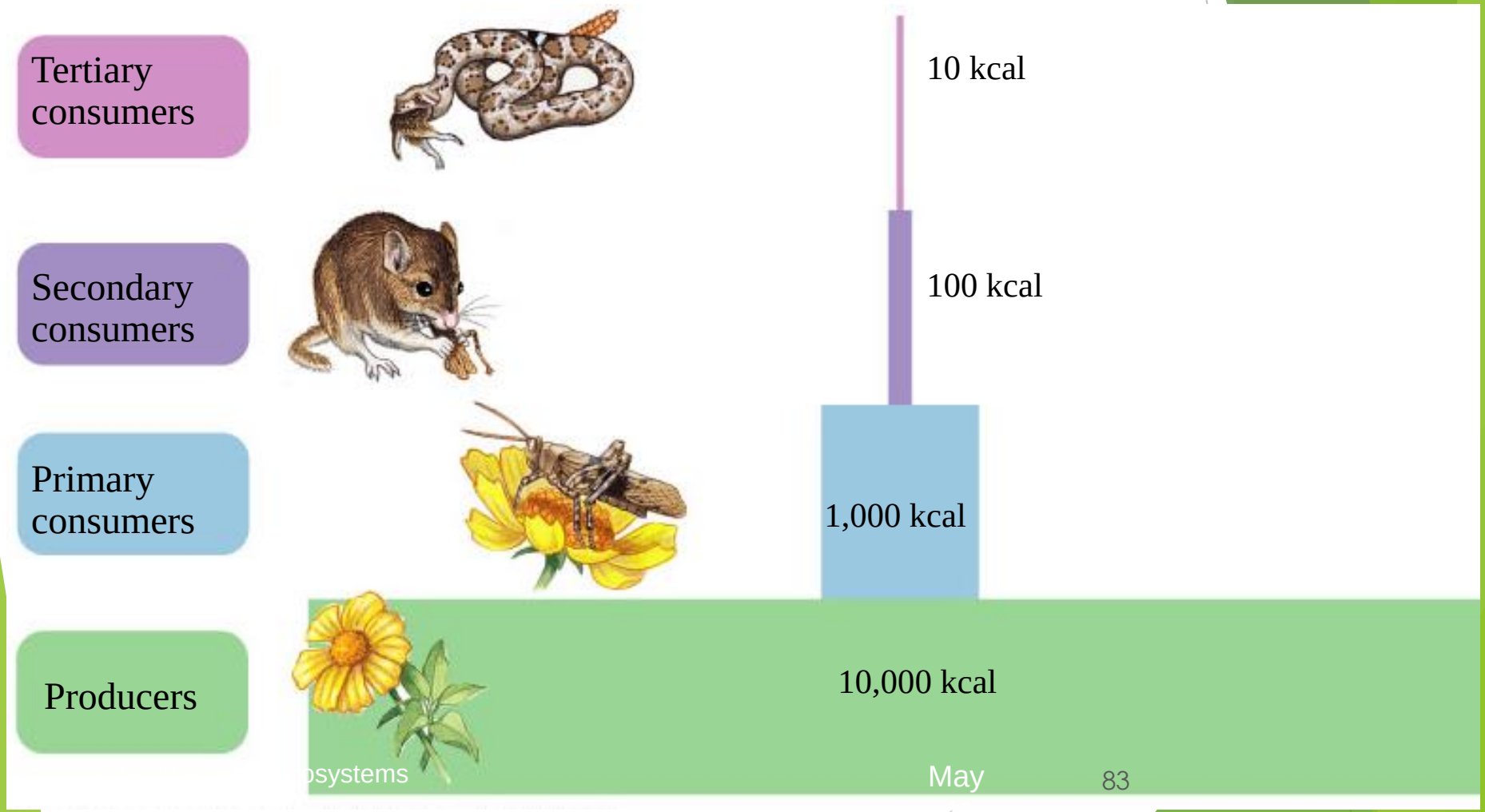


Pyramids of Energy

- On average, about 10 percent of the energy available within one trophic level is transferred to the next trophic level (trophic efficiency)
- The more levels that exist between a producer and a consumer, the smaller the percentage of the original energy from producers that is available to that consumer.



Pyramids of Energy



Pyramids of energy

- Advantages
  - Takes into account the PRODUCTIVITY
  - Addresses the fact that weight for weight, two species do not necessarily have the same energy content
- Disadvantages
  - Very difficult to obtain data
  - Destructive
  - Problems identifying trophic level

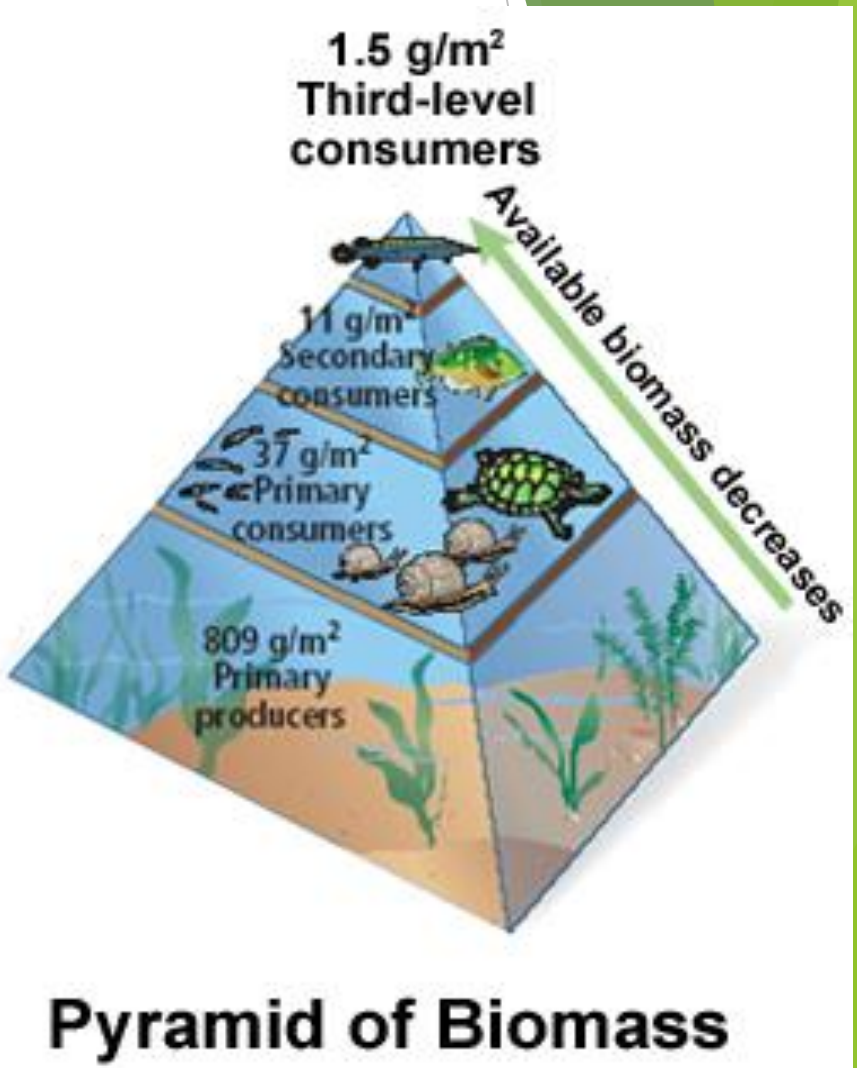
2. Pyramids of Biomass

- The total amount of living tissue within a given trophic level is called its **biomass.**
- The amount of biomass a given trophic level can support is determined, in part, by the amount of energy available.



2. Pyramids of Biomass

- Illustrates the relative amount of living organic matter at each trophic level.
- Indicates the total dry mass of the organisms in each trophic level
- Typically, the greatest biomass is at the base of the pyramid, as is seen in the field ecosystem modeled here.



Pyramid of Biomass

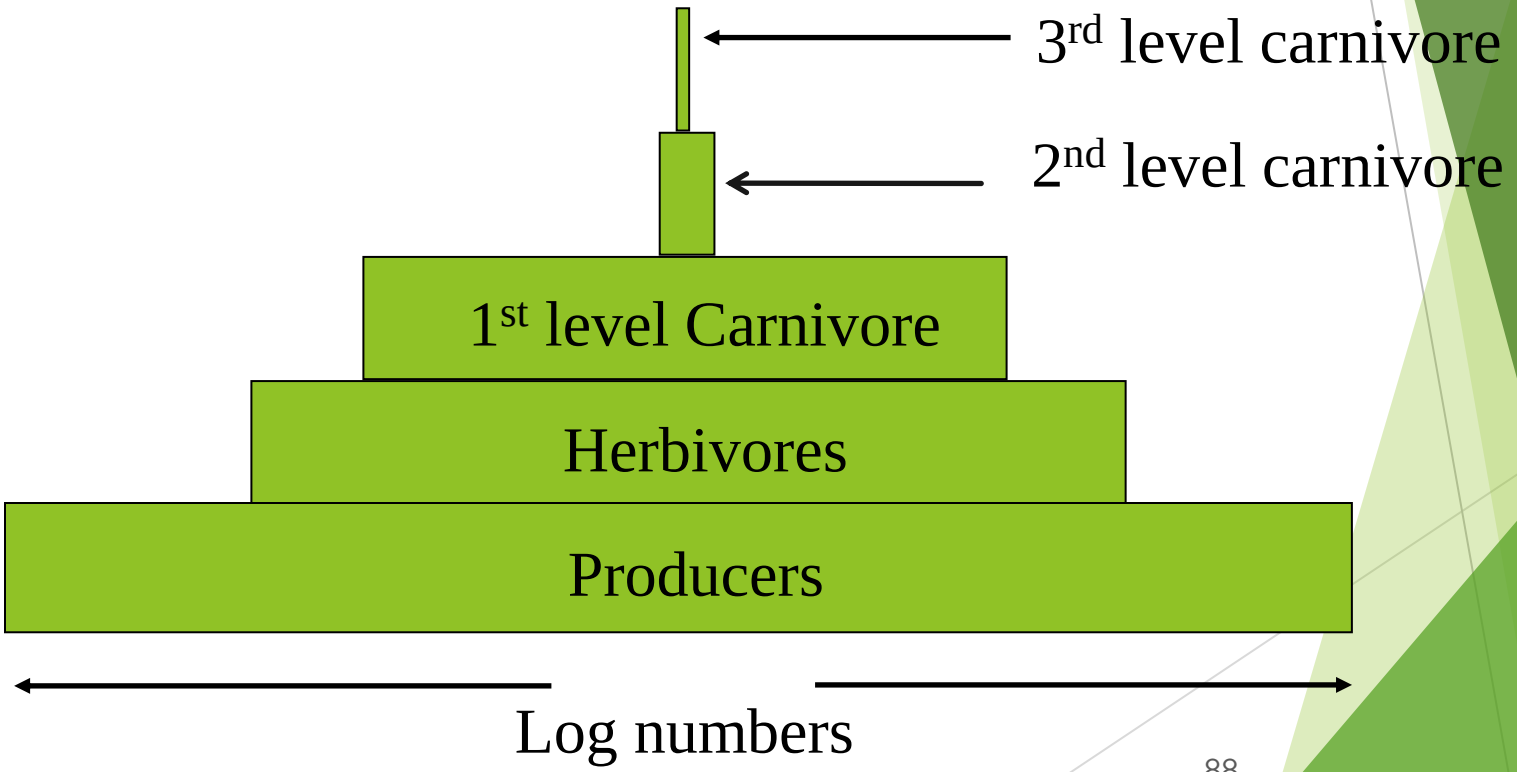
Pyramids of Biomass

- How might this data be collected?
  a) Individual organisms weighed and counted.
  b) Dry mass should be compared to eliminate errors due to water content

- Advantages
  - The data is more accurate.
  - Eliminates misleading problems of producer size differences
- Disadvantages
  - Laborious and expensive
  - Destructive
  - Can also be misleading in some circumstances

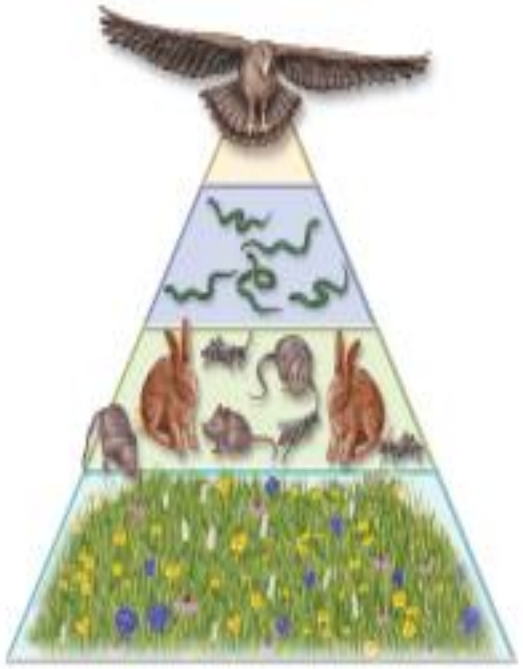
3. Pyramids of Numbers

- A pyramid of numbers shows the relative number of individual organisms at each trophic level in an ecosystem.



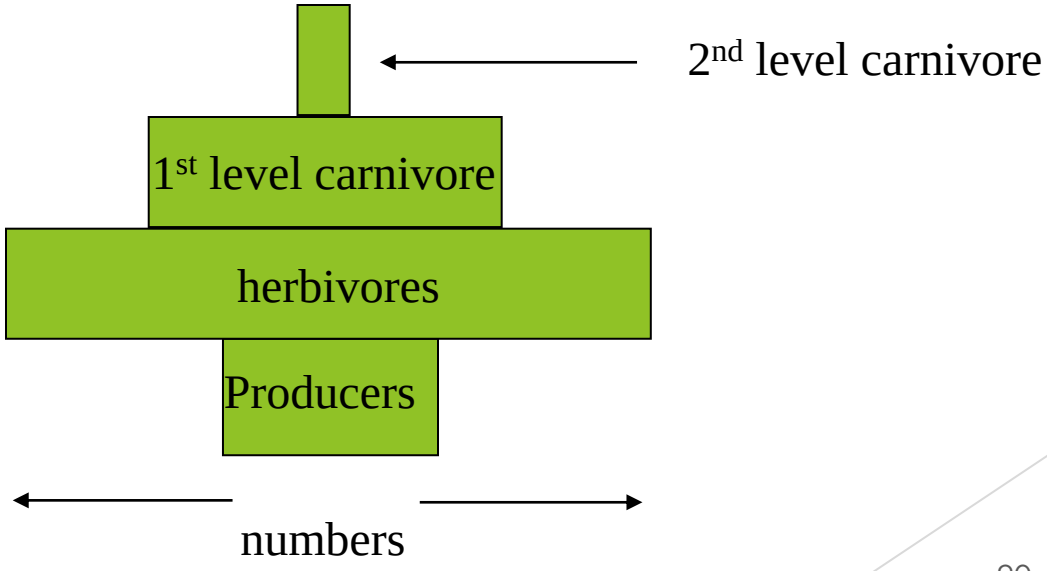
3. Pyramids of Numbers

In most ecosystems, the shape of the pyramid of numbers is similar to the shape of the pyramid of biomass for the same ecosystem, with the numbers of individuals on each level decreasing from the level below it.



3. Pyramids of Numbers

- In some cases, however, consumers are much smaller than organisms they feed upon.
- Thousands of insects may graze on a single tree, for example. The tree has a lot of biomass, but represents only one organism.
- In such cases, the pyramid of numbers may be turned upside down, but the pyramid of biomass usually still has the normal orientation.



Pyramids of Number

- Advantage
  - Data relatively easy to collect using simple sampling
- Disadvantages
  - Producers vary in size
  - Large range of numbers
  - Trophic level difficult to work out